

Generalizing across Networks: Evaluating Model Transferability for Intrusion Detection

Miguel Silva^a, João Vitorino^b, Daniela Pinto^c, Ivone Amorim^d, Eva Maia^e and Isabel Praça^f

GECAD, ISEP, Polytechnic of Porto, rua Dr. António Bernardino de Almeida, 4249-015 Porto, Portugal

Keywords: Cybersecurity, Deep Learning, Feature Selection, Machine Learning, Model Generalization.

Abstract: As cyber threats continue to evolve, attackers improve their techniques to optimize the effectiveness of their malicious activities. In response, there has been a significant increase in the implementation of Machine Learning (ML) and Deep Learning (DL) in Network Intrusion Detection Systems (NIDS). Choosing high-quality, representative datasets is a critical step in improving detection accuracy; however, many datasets have been available for a long time, some for over two decades, raising questions about the ability of ML and DL models to generalize across different and possibly outdated network scenarios. This study evaluates the generalization capability of models such as LGBM, RF, XGB, and LSTM, particularly in identifying previously unseen attacks. A significant challenge addressed is the inconsistency introduced by different flow exporters, which compute features differently, affecting model reliability. To mitigate this, HERA is used to rebuild datasets directly from original PCAP files. Additionally, we investigated the impact of feature selection on generalization and examined how performance changes when combining different datasets. When evaluated on alternative datasets, however, their behavior diverges, and in some cases, similar traffic patterns allow for reasonable generalization, while in others, the uniqueness of the dataset leads to significant performance drops.

1 INTRODUCTION

The rise in complexity and frequency of cybersecurity threats has made NIDS a necessity for protecting organizations and their assets. These systems perform a critical role in identifying malicious or anomalous network traffic, proving to be invaluable during cyber attacks (Zha et al., 2025). Conventional rule-based methodologies are often ineffective against novel threats, as attackers can bypass signature-based systems by altering known patterns or deploying previously unseen tactics. In this context, Artificial Intelligence (AI) has gained popularity due to its ability to detect threats by recognizing patterns in real-time traffic and identifying similarities to known malicious behaviors (Rai et al., 2025).

Nevertheless, the effectiveness of AI-based models largely depends on the quality and relevance of their training data (Gong et al., 2023). These models

rely on subtle variations in network behavior to accurately classify traffic, which requires the use of comprehensive datasets that represent both benign and malicious behaviors.

Despite advances in this domain, many existing studies still rely on outdated datasets that may not accurately reflect modern traffic behaviors or attack techniques. Although these models might perform well on their training datasets, their ability to generalize and maintain their performance in new scenarios remains unexplored. Moreover, as several datasets claim to represent similar network scenarios, they are often created using different tools, methodologies, or network topologies, leading to inconsistencies that complicate cross-dataset comparisons. Additionally, since traffic statistics are extracted using different flow exporters, the same packet capture may produce different feature values, depending on the exporter used. This variability raises concerns about the reliability and compatibility of features, as well as the validity of model evaluations across different datasets.

This study evaluates the generalization capabilities of several ML and DL models. For this evaluation, the models are trained and tested across multiple

^a  <https://orcid.org/0009-0008-6630-9939>

^b  <https://orcid.org/0000-0002-4968-3653>

^c  <https://orcid.org/0009-0000-3003-6694>

^d  <https://orcid.org/0000-0001-6102-6165>

^e  <https://orcid.org/0000-0002-8075-531X>

^f  <https://orcid.org/0000-0002-2519-9859>

datasets, all of which are processed through the same flow exporter HERA (Pinto et al., 2024), to extract flow-level features from packet capture data, ensuring consistency in feature values. Moreover, this study assesses whether these models can maintain their performance across different datasets and the impact of reducing the number of features on their generalization capabilities.

This present paper is organized into the following sections: Section 2 provides an overview of previous studies on the importance of using current datasets and identifies challenges associated with different flow exporter tools. Section 3 details the datasets, feature selection methods, and ML and DL models employed. Section 4 presents and discusses the experimental results, with a focus on model generalization across datasets. Section 5 addresses the main conclusions and outlines future research directions.

2 RELATED WORK

The growing amount of cyber threats has led organizations to adopt intelligent NIDS solutions, specifically ML and DL, to detect anomalies and cyber attacks in network traffic. As each model can vary in complexity and behavior, and these models require initial information to train them, ML engineers use known NIDS datasets to evaluate and select the appropriate model for their organization. Recent studies (Boukraa et al., 2025; More et al., 2024; Tariq et al., 2025) show that models can achieve high levels of performance on publicly available datasets, being mostly higher than 96% F1-score using standard ML models such as Decision Tree, Random Forest, and Support Vector Machine on datasets such as UNSW-NB15. Regarding other known datasets, in CICIDS2017, ML models achieve comparable results, yielding high F1-scores, even though different models, namely, Naive Bayes, achieve lower than 50% F1-score. A more recent dataset, HIKARI2021, also achieves very high results, ranging the 88% to 96% F1-score, although similar to CICIDS2017, the Gaussian Naive Bayes performs the worst, but with HIKARI2021, it manages to hit 79% F1-score. Overall, ML ensembles, such as LGBM, RF, and XGB, approaches for NIDS tend to outperform individual models, achieving higher efficiency and more robust evaluations (reddy and Sathya, 2022; Biswas et al., 2024; Khajgiwale, 2024). Additionally, DL models, namely LSTM, have recently gained attention due to their ability to accurately classify network flows by capturing temporal patterns in network traffic (Vibhute et al., 2024).

Since raw network data are not usable as ML models' input, flow exporters are required to convert packet-level data into structured feature representations. As there are multiple available flow exporters, any inconsistencies created during the feature extraction process can negatively impact model performance and even lead to unreliable detection outcomes (Pinto et al., 2025). In addition to traffic statistics, a key limitation of many NIDS datasets is structural inconsistencies, namely, a lack of a common feature set (Sarhan et al., 2022). This makes cross-dataset evaluations more difficult and hinders model transferability.

As the importance of high quality training data deeply affects the behavior and results obtained from models, specifically when tested in production environments, a known problem with some NIDS datasets is that they have been in use for over 25 years, with some of the earliest examples dating back to the late 1990s. Despite their historical significance, these datasets are frequently subjected to criticism for being outdated (Alshaibi et al., 2022), largely due to a lack of continued maintenance or updates by their original authors. Nevertheless, they are still commonly used in research to evaluate the performance and effectiveness of NIDS (Dixit, 2025; Wang et al., 2025). While these datasets may produce strong results in controlled experiments, they often fail to reflect current traffic patterns, protocols, or modern threat vectors, limiting their real-world applicability (Hnamte and Hussain, 2021; Kenyon et al., 2020).

In summary, previous studies have highlighted the structural inconsistencies and declining relevance of commonly used NIDS datasets. Even though other recent studies have presented high performance results using these datasets, these achieve such results in their respective testing sets. Therefore, to address these issues, we evaluate the generalization capability of ML models across multiple network traffic scenarios to more clearly evaluate the models' application to unseen traffic information by making the models predict traffic flows from other datasets.

3 METHODOLOGY

This section describes the considered datasets, feature selection methods, and models. This study was carried out on a standard laptop with 16GB of RAM, a 6-core CPU, and an 8GB GPU. The implementation was developed with Python and the following libraries: *numpy* and *pandas* for general data manipulation, *xgboost* for the implementation of XGB, *lightgbm* for LGBM, *tensorflow* for LSTM, and *scikit-learn* for both RF and feature selection.

3.1 Datasets

The datasets used in this study contain labeled, flow-level representations of network traffic, including normal traffic and network attacks. Although the primary objective is binary classification, some datasets provide granular attack type labels. To ensure feature consistency, we processed the original PCAP files from all datasets using HERA to generate standardized 60-second flow statistics.

The GeNIS dataset (Silva et al., 2025a), created in 2025, includes sequential network attack scenarios, such as denial-of-service, brute-force, and reconnaissance. HIKARI-2021 (Ferriyan et al., 2021) contains a mix of brute-force, crypto-miner, and probe cyber attacks. While both datasets include brute-force attacks, they differ in protocol usage, with GeNIS using FTP, SMB, and SSH protocols and HIKARI-2021 using XML and HTTPS. These differences establish a diverse testing environment for evaluating model generalization.

CICIDS2017 (Sharafaldin et al., 2018), a widely adopted benchmark dataset, contains a variety of attacks, such as FTP and SSH brute-force, and denial-of-service. While this dataset and GeNIS share some attacks, namely brute-force attacks, GeNIS uses hydra, whereas CICIDS2017 uses a different tool, patator.

The UNSW-NB15 (Moustafa and Slay, 2015), which has been published for over a decade, is also the dataset with the most different types of attacks, totaling nine unique ones. Since this dataset contains attacks carried out several years ago, it is useful for understanding how attack and benign behavior have changed over the past ten years and whether models trained with current attacks and normal traffic can identify past attacks.

Before the datasets could be used, a data pre-processing phase was required. This phase involved creating a stratified training set and a holdout set with 70% and 30% of the data, respectively. Since the datasets differ by ten years in age, categorical features such as protocols, services, and ports were disregarded to ensure that the prediction of the models' prediction is solely based on traffic behavior. Additionally, features containing more than 10% missing values were also removed across all datasets. Then, rows with missing data were removed. The final set of features included 40 features, excluding the label. The dataset with the most attack flows was GeNIS, although it also had the least amount of benign traffic. Table 1 contains the number of flows per class in each set.

3.2 Feature Selection

To select the most meaningful and most representative features for each dataset, five different feature selection methods were employed. The values obtained from each method were scaled and combined into a single value, resulting in an ordered list of the most relevant features for each dataset.

These methods were **Recursive Feature Elimination**, which using a standard RF model, recursively reduced the feature set by removing the less important features one by one until the most impactful feature is reached (Fuhnwi et al., 2024); **Chi-Squared Test**, which measures the dependence between each feature and label to identify features with stronger relationships (Reddy et al., 2024); **Mean Absolute Deviation**, which measures the dispersion in a feature, as an alternative to standard deviation, measuring the distance of the average data points from the mean (Silva et al., 2025b); **Dispersion Ratio**, which results in the variability among features to ensure they carry enough information for models to understand how the values are used to make predictions (Setiawan et al., 2024); **Information Gain**, a method that measures the change in entropy, specifically the decrease in overall information, when a feature is removed, resulting in the relevance of that feature (Gao et al., 2024).

3.3 Models and Fine-Tuning

Three ML tree-based models (LGBM, RF, and XGB) and a Deep Learning model (LSTM) were selected. The optimal configuration for each ML model was obtained through a grid search involving the size and learning rate of each model, for the full and feature-selected set of features. The best configurations were selected through a 5-fold cross-validation, with evaluation based on the F1-score. The models and their fine-tuned hyperparameters are described below.

Light Gradient Boost Machine. LGBM (Ke et al., 2017) is an efficient ensemble learning framework that uses gradient boosting to build strong learners by adding weak learners in a gradient descent manner. The configuration of this model is summarized in Table 2.

Random Forest. RF (Breiman, 2001) is an algorithm based on decision trees to improve prediction accuracy. Each tree makes a prediction, and the total predictions are grouped by majority to reach the final prediction. Table 3 summarizes the configuration.

Extreme Gradient Boosting. XGB (Chen and Guestrin, 2016) is a gradient boosting learning method that combines the performance of multiple

Table 1: Benign and malicious traffic rows in training and testing sets.

Dataset	Training set		Testing set	
	Benign	Malicious	Benign	Malicious
GeNIS	19,005	238,984	8,145	102,422
HIKARI-2021	258,467	50,859	110,772	21,797
CICIDS2017	792,461	129,141	339,627	55,346
UNSW-NB15	674,310	31,844	288,991	13,647

Table 2: Summary of LGBM configuration.

Parameter	Value
Maximum leaves in tree	32
Minimum loss reduction	0.01
Minimum samples in leaf	16
Learning rate	0.1 to 0.3
Maximum tree depth	8 or 16
Number of estimators	100

Table 3: Summary of RF configuration.

Parameter	Value
Criterion	Gini Impurity
Number of estimators	100
Maximum features	$\sqrt{\text{Number of features}}$ or <i>None</i>
Maximum tree depth	8 or 16

decision trees to improve the performance of a model, with each tree trained to correct the errors made by the previous models. To choose the best node splits, the Histogram method was used. The configuration is summarized in Table 4.

Table 4: Summary of XGB configuration.

Parameter	Value
Learning rate	0.1 to 0.3
Minimum loss reduction	0.01
Number of estimators	100
Maximum tree depth	8 or 16

Instead of an automated hyperparameter search, the model architecture and configuration were manually adjusted based on the obtained experimental results to train the LSTM. Early stopping was employed during training to prevent overfitting and automatically restore the weights from the epoch with the best performance. All input features were normalized using standard scaling, and the model was trained using 80% of each training set, with the remaining 20% used for validation.

Long Short-Term Memory. LSTM (Hochreiter and Schmidhuber, 1997) is a type of Recurrent Neural Network designed to capture long-term dependencies in sequential data. Since the data in these datasets are not sequences, each sample is treated as a one-step sequence, enabling the LSTM to capture nonlinear interactions among features. This network's configuration is summarized in Table 5. The network has one input, one output layer, and two hidden layers: an LSTM layer with 64 units and a Dense layer with 32 units.

Table 5: Summary of LSTM configuration.

Parameter	Value
Dropout	0.2
Optimizer	Adam
Learning rate	0.001
Batch size	32
Epochs	30
Early stopping patience	3

4 RESULTS

This section presents and discusses the results obtained from the feature selection method, as well as the model evaluations. The ten most relevant features, with the highest rankings for each dataset, were selected for comparison of the models' generalization capacity with all the available features and a selected subset.

The evaluation metrics used to compare and assess the generalization of models were the F1-score (F1S) and False Positive Rate (FPR). Since class imbalance was evident in each dataset (Table 1), Accuracy was excluded as models could achieve high Accuracy by classifying all samples as benign, particularly in the case of HIKARI-2021, CICIDS2017, and UNSW-NB15, which have more benign flows than malicious. Therefore, F1S was chosen since it provides the most comprehensive view of a model's generalization, and FPR was also selected to determine whether models classify most of the testing samples

Table 6: Top ten features selected per dataset.

GeNIS	HIKARI-2021	CICIDS2017	UNSW-NB15
Offset	Load	Offset	Offset
SrcLoad	DstLoad	Load	SrcLoad
Load	SrcLoad	DstLoad	SrcLoss
DstLoad	Offset	SrcLoad	<u>Loss</u>
<u>Ssaddr</u>	DAppBytes	<u>SrcBytes</u>	Load
DAppBytes	Dur	SAppBytes	<u>Sdaddr</u>
Rate	Sum	<u>TotBytes</u>	<u>DstLoad</u>
RunTime	RunTime	<u>TotAppByte</u>	Rate
SrcRate	<u>Min</u>	<u>dMaxPktSz</u>	<u>Seq</u>
Sum	SAppBytes	Dur	SrcRate

as benign. Optimal results would be 100% of F1S and 0% of FPR.

4.1 Analysis of Feature Selection Results

Regarding the most relevant features for each dataset, the top ten features represented from 58% to 69% of the total information, which were identified by the methods using only a fourth of the total available features. The four most impactful features for GeNIS, UNSW-NB15, and HIKARI-2021 are the same, although their order changes. These features are also present in the overall top ten of CICIDS2017, however, they are ranked lower. HIKARI-2021 and CICIDS2017 datasets show the most unrelated features among the other datasets, with four unique features, while GeNIS and UNSW-NB15 have nine features in common with the other datasets, but two are not shared between them: Rate and Sum for GeNIS, and SAppBytes and Dur for UNSW-NB15. The results of the feature selection methods are presented in Table 6, where the unmatched features are underlined.

The results of the analysis resulted in a final set of 21 unique features. These features were selected from all the datasets and used as input for the experiments with the reduced feature set.

4.2 Performance on Full Feature Set

After the pre-processing stage, all the ML models achieved high performance when evaluated on the respective testing set, with F1S always higher than 90%, and with a low FPR. Of these, the UNSW-NB15 dataset had the worst scores, while all the other datasets had at least one model with perfect scores. The LSTM model consistently performed the worst, with a difference in scores of more than 10% on the UNSW-NB15 dataset.

Regarding the models' ability to generalize and predict from other datasets, no ML model trained on GeNIS could accurately classify the flows from the UNSW-NB15 dataset. The models predicted all samples as benign, achieving 0 F1S and FPR. However, the LSTM did not exhibit the same behavior, although it was also unable to achieve a good score in either metric. When predicting flows from other datasets, RF showed better generalization, achieving 67% and 86% F1S on HIKARI-2021 and CICIDS2017, respectively, and achieving a much lower FPR than the other models subjected to the same data. Notably, both of these datasets have a much larger quantity of benign traffic than GeNIS, and this model misclassified 17,861 and 17,642 instances of benign traffic as malicious.

When trained with the HIKARI-2021 dataset, no ML model achieved good performance on other datasets. Most models achieved a perfect FPR, suggesting that they mostly classified all the samples as benign, but did not identify any attacks. The LSTM, on the other hand, was able to achieve a high F1S on the CICIDS2017 dataset and maintain a low FPR, classifying 9,867 benign flows as attacks.

The models trained in the CICIDS2017 dataset took the longest time to train and were incapable of generalizing the information learned, rendering them useless for predicting flows from other datasets. Across the GeNIS and HIKARI-2021 datasets, these models classified most of the samples as benign, as indicated by an FPR lower than 10%. However, when evaluating the UNSW-NB15 dataset, the models behaved differently than those trained on other datasets, achieving a very high FPR, except for LGBM, which nearly reached the 50% mark on F1S. This suggests that most samples evaluated were classified as attacks.

Although the RF and LGBM were trained on the UNSW-NB15 and can identify attacks in GeNIS, they also misclassified a large portion of benign samples as attacks. Since the GeNIS dataset consists mostly of attack samples, it is relevant to note that 1% of

FPR corresponds to approximately 82 flows. The LGBM model performs worse when presented with the HIKARI-2021 testing set, while the XGB model achieves a score similar to the RF model in both F1S and FPR. Neither ML nor DL models can successfully detect attacks within the CICIDS2017 testing set. This aligns with the results obtained using the CICIDS2017 as the training set, as it classified most of the flows from UNSW-NB15 as attacks, and when trained on UNSW-NB15, the models classified most of the flows from CICIDS2017 as benign. Table 7 provides the models' results, highlighting the evaluation obtained for each model with its respective testing set.

4.3 Performance on Reduced Feature Set

Using approximately half of the available features, all models trained on the respective testing set, except RF trained on UNSW-NB15, showed a slight decrease in F1S and an increase in FPR. All ML models were able to finish their training faster, mainly due to the lesser amount of information provided to them, except for RF and XGB trained on CICIDS2017, and RF trained on UNSW-NB15. The LSTM network took significantly more time to finish training in three out of four datasets, potentially due to the early stopping patience value that allowed the model to continue training through additional epochs to achieve lower validation loss.

Among the models trained on the GeNIS dataset, only the RF achieved good F1S and FPR results on the HIKARI-2021 testing set. The other models did not generalize the dataset well enough to accurately predict the class of each flow, as they achieved a high FPR, indicating that a large amount of the benign flows were classified as attacks. The exception was UNSW-NB15, where all the ML models classified the flows as mostly benign.

While trained on HIKARI-2021, the LSTM model performed well in predicting attacks on the GeNIS dataset while maintaining a low FPR. All models achieved an F1S greater than 50% on the CICIDS2017 dataset, especially the RF, which achieved an F1S of almost 94% and an FPR of less than 1%. This represents a significant performance improvement compared to the model fitted with all features, which achieved an F1S of less than 1%. However, none of these models could identify attacks on the UNSW-NB15 dataset, with most flows labeled as benign traffic.

Using this smaller set of features, the XGB trained on the CICIDS2017 dataset, which had previously

been unable to successfully detect attacks on other testing datasets, achieved an F1S of 76% while maintaining a perfect FPR score. No other ML or DL model trained on this dataset achieved such high evaluation performance on any other dataset with all features.

After being trained on UNSW-NB15, RF and LSTM achieved an F1S greater than 50% on the GeNIS dataset, although they also achieved a high FPR. XGB, on the other hand, achieved a similar F1S but a much lower FPR on the CICIDS2017 testing set. Table 8 provides the results of the models and highlights the evaluation obtained for each model with its respective testing set.

4.4 Performance of Combination on Full Feature Set

Models trained on a single dataset performed poorly when classifying flows from other datasets, which highlights the significant variability in traffic characteristics. To improve generalization, we trained models on all possible combinations of three out of the four available datasets, using the remaining dataset as an unseen test set. By doing so, the models received the most diverse training data possible while being able to be evaluated on an entirely new dataset. The detailed results of these experiments are presented in Table 9.

When evaluated using the complete set of features, the models had difficulty detecting attacks in the GeNIS dataset. The simpler tree-based RF model was the only one to achieve an F1S above 50%, reaching 66%. FPR scores from these models were generally low or negligible, suggesting that these, particularly ones other than RF, misclassified a significant number of GeNIS attacks as benign, similar to flows observed in other datasets. A similar situation occurred with CICIDS-2017 when it was excluded from training. RF achieved an F1S of 61%, while the LSTM surpassed 50% only in this specific case. However, unlike GeNIS, CICIDS-2017 showed distinctly higher FPR scores ranging from 19% to 26% for RF and LSTM.

By contrast, when HIKARI-2021 was excluded from the combination training set, tree-based models such as LGBM and XGB performed well, achieving F1S around 85%. This suggests that HIKARI-2021 traffic is similar to that of other datasets, enabling models trained on combined data to generalize more effectively.

Performance on UNSW-NB15, however, was notably different. Across all models except LSTM, results were poor despite low FPR values. This sug-

Table 7: Results obtained using all features.

Training Set	Model	Test Set								Training Time (s)
		GeNIS		HIKARI		CICIDS		UNSW		
		F1S	FPR	F1S	FPR	F1S	FPR	F1S	FPR	
GeNIS	LGBM	100	0	31.12	70.72	35.05	57.46	0	0	2.91
	RF	100	0	67.95	16.12	86.21	5.19	0	0	14.69
	XGB	100	0	30.81	76.10	35.40	57.75	0	0	2.44
	LSTM	99.97	0.16	0.0	52.15	50.02	32.24	13.56	58.86	67.06
HIKARI	LGBM	0.03	0	99.99	0	0.41	0	0	0	2.07
	RF	0	0	100	0	0.02	0	0	0	12.34
	XGB	1.44	10.85	100	0	0.28	0.10	0	0	1.31
	LSTM	6.50	0	99.84	0.04	86.59	2.90	33.17	0.52	155.69
CICIDS	LGBM	14.91	0.02	0	0.32	99.98	0	48.02	0.70	6.50
	RF	14.97	3.54	0	7.45	100	0	3.29	97.76	143.15
	XGB	14.98	3.03	0	0.64	99.99	0	10.47	80.24	4.18
	LSTM	15.01	0.33	0.01	8.49	99.96	0.01	9.60	88.21	424.28
UNSW	LGBM	64.82	77.62	27.26	54.66	15.76	15.34	91.95	0.37	4.04
	RF	65.72	39.48	51.07	13.28	2.02	11.78	91.88	0.36	82.59
	XGB	23.12	41.55	50.85	15.85	20.70	11.79	93.57	0.33	8.49
	LSTM	42.77	24.60	1.02	6.97	2.98	12.88	79.65	2.41	319.95

Table 8: Results obtained using the reduced feature set.

Training Set	Model	Test Set								Training Time (s)
		GeNIS		HIKARI		CICIDS		UNSW		
		F1S	FPR	F1S	FPR	F1S	FPR	F1S	FPR	
GeNIS	LGBM	100	0.01	29.12	78.05	35.56	57.71	0	0	1.47
	RF	100	0	67.91	16.17	46.04	38.19	0	0	8.78
	XGB	100	0	30.77	78.86	35.38	59.50	0.04	0	0.73
	LSTM	99.95	0.70	29.30	75.22	36.45	54.01	0.85	69.40	180.09
HIKARI	LGBM	0.13	0	99.99	0	58.07	19.22	0	0	1.40
	RF	0.78	0	99.99	0	93.77	0.15	0	0	10.17
	XGB	41.25	0.08	99.99	0	57.75	19.54	0	0	0.92
	LSTM	70.50	2.09	99.41	0.21	77.90	6.12	0	0	139.19
CICIDS	LGBM	27.73	0	9.03	2.82	99.99	0	0	0	3.98
	RF	34.78	0	6.55	3.24	99.99	0	0	0	220.66
	XGB	75.90	0	6.56	2.94	99.99	0	0	0	4.71
	LSTM	1.38	0	0.01	0.57	99.96	0.01	0.12	0.04	916.21
UNSW	LGBM	16.00	46.42	45.49	16.72	18.53	5.18	91.56	0.38	2.68
	RF	51.53	26.84	35.88	61.40	10.16	38.30	92.32	0.33	357.64
	XGB	4.37	8.66	32.36	5.31	51.40	4.32	93.65	0.32	6.50
	LSTM	54.28	55.40	14.18	29.94	5.04	39.55	72.47	3.45	639.10

gests that the attack patterns found in the UNSW-NB15 dataset differ substantially from those found in other datasets. Consequently, the models incorrectly classified many attack flows as normal activity.

Including GeNIS in the training set changed the outcome considerably as models became highly effective at distinguishing attacks from benign flows. They achieved over 99% F1S with an extremely low FPR of no more than 0.7% on LSTM. A similar trend was observed with CICIDS-2017. Results also improved substantially when HIKARI-2021 was included in the training combinations. In particular, the LSTM increased its F1S by more than fivefold while reducing the FPR to less than a tenth across all combinations. By contrast, UNSW-NB15 produced the weakest results consistently. Its uniqueness meant that, whether used in isolation or combined with others, models relied heavily on patterns specific to UNSW-NB15, limiting their ability to generalize to other datasets.

4.5 Performance of Combination on Reduced Feature Set

Using a reduced subset of features caused the models to behave differently when classifying unseen datasets compared to those included in the training set, as can be seen in Table 10. For instance, on GeNIS, tree-based models such as LGBM and XGB achieved notable F1S improvements. In contrast, RF, which had previously achieved an F1S of 66%, dropped drastically to 6%, rendering it practically ineffective. On HIKARI-2021, LGBM and XGB produced comparable gains, though at the cost of doubling their FPR. Meanwhile, RF and LSTM diverged sharply in performance, with RF losing 48% of its F1S and LSTM improving by 42%.

Unlike the results obtained with the full feature set, all tree-based models on CICIDS-2017 surpassed 50% F1S. However, this improvement came with higher FPR values, averaging approximately 20%. Meanwhile, LSTM showed a substantial decline, doubling its FPR and falling below 50% F1S.

As before, UNSW-NB15 showed its unique properties, as models trained on other datasets could not accurately classify its flows, further emphasizing the dataset's distinctiveness.

Including GeNIS in the training combinations once again resulted in nearly perfect performance on its testing set. However, the LSTM achieved slightly lower F1S and higher FPR scores compared to the full feature set. Similarly, testing on HIKARI-2021 and CICIDS-2017 produced consistently high F1S and negligible FPR scores across models. Results on UNSW-NB15 remained weaker, nonetheless, with

lower F1 and higher FPR. All models showed reduced performance, with lower F1S and higher FPR, with the exception of XGB in combinations excluding GeNIS or CICIDS-2017. Although the F1S remained above 50%, they ranged widely from 65% to 94%, which is lower than the values observed in the other testing sets.

5 CONCLUSION

This study evaluated the generalization capacity of different ML and DL models to successfully detect attacks on unseen data, from different datasets, while also testing their compatibility given the large age gaps between them. To accomplish this, the information from each PCAP file in the GeNIS, HIKARI-2021, CICIDS2017, and UNSW-NB15 datasets were extracted using the HERA flow exporter tool. The same pre-processing was applied to all datasets, resulting in a set of 40 unique features.

A reduced subset of unique values was identified through a combination of several feature selection methods and used for comparison with the full set. The results show that, when evaluated using their respective test sets, the models achieve very good performance using the full set of features and similar results using half the features, requiring mostly lower training times.

Overall, the models generally struggled with comprehensive generalization across all testing sets, particularly when classifying flows from UNSW-NB15. However, the RF trained on GeNIS, achieved a high F1S on three out of four datasets using all available features. More notably, the LSTM trained on HIKARI-2021 demonstrated superior generalization using a reduced feature set, with an F1S of over 70% on three out of four datasets. This subset proved to be more compatible with the HIKARI-2021 trained models, which performed much better than the full set.

Several combinations of datasets were also made to study the effects of adding more information and behavioral patterns on the results obtained from the models. The results show that the models become very capable of successfully predicting flows from a test set when their respective training set is included, and their predictions change as they have more information from other datasets. Specifically, the combination without the HIKARI-2021 in training, using all features and a smaller subset, achieved higher scores than any individual model on the HIKARI-2021 testing set. While some models improve with these combinations, others do not. This indicates that

Table 9: Results obtained from dataset combination using all features.

Without Training Set	Model	Test Set								Training Time (s)
		GeNIS		HIKARI		CICIDS		UNSW		
		F1S	FPR	F1S	FPR	F1S	FPR	F1S	FPR	
GeNIS	LGBM	16.61	0.33	99.98	0.01	99.98	0.01	89.73	0.50	11.15
	RF	66.34	0	99.99	0	99.99	0	92.37	0.36	840.41
	XGB	0	0	100	0	100	0	89.73	0.34	20.08
	LSTM	14.80	0	99.66	0.07	99.94	0.02	79.49	2.42	581.73
HIKARI	LGBM	100	0	85.50	2.63	99.99	0	92.04	0.37	9.75
	RF	100	0	64.27	1.94	99.99	0	91.32	0.37	1448.36
	XGB	100	0	84.91	2.90	99.98	0	93.13	0.32	15.11
	LSTM	99.96	0.70	18.98	9.65	99.92	0.02	79.62	2.32	573.13
CICIDS	LGBM	100	0.01	100	0	10.44	2.25	91.19	0.42	7.36
	RF	99.95	0.09	99.96	0.01	60.74	18.59	90.85	0.40	1015.25
	XGB	100	0	99.98	0.01	23.44	0.68	92.36	0.37	9.60
	LSTM	99.98	0.22	99.68	0.10	54.09	26.25	79.68	2.38	326.18
UNSW	LGBM	100	0	99.99	0	99.99	0	0.14	0.4	8.79
	RF	99.98	0.59	99.85	0.06	99.97	0.01	0.59	0.51	1008.33
	XGB	100	0	99.99	0	99.99	0	0	0.13	9.23
	LSTM	99.97	0.16	99.80	0.05	99.96	0.01	22.14	15.43	975.191

Table 10: Results obtained from dataset combination using a subset of features.

Without Training Set	Model	Test Set								Training Time (s)
		GeNIS		HIKARI		CICIDS		UNSW		
		F1S	FPR	F1S	FPR	F1S	FPR	F1S	FPR	
GeNIS	LGBM	65.77	0.43	99.94	0.02	99.92	0.02	86.85	0.64	7.72
	RF	5.79	1.87	99.99	0	99.98	0.01	90.60	0.43	886.50
	XGB	67.05	0.21	99.99	0	99.98	0.01	93.38	0.32	16.93
	LSTM	19.94	5.13	98.22	0.68	98.80	0.39	73.10	3.09	1730.74
HIKARI	LGBM	100	0	85.17	5.46	99.99	0	90.03	0.43	7.89
	RF	100	0	16.37	0.41	100	0	88.83	0.49	188.84
	XGB	100	0	82.45	5.11	99.9	0	92.34	0.35	11.67
	LSTM	99.92	1.28	60.93	7.02	99.51	0.16	65.45	2.64	945.31
CICIDS	LGBM	100	0	99.99	0	61.73	19.10	89.84	0.46	4.91
	RF	100	0	99.98	0	58.55	20.19	90.72	0.42	519.08
	XGB	100	0	99.98	0	55.99	19.54	93.68	0.32	11.05
	LSTM	99.92	1.22	99.30	0.19	40.80	44.74	75.52	2.77	1347.10
UNSW	LGBM	100	0	99.99	0	99.99	0	0	0	5.98
	RF	100	0	99.99	0	99.99	0	3.42	0.26	483.31
	XGB	100	0	99.99	0	99.98	0	0	0	7.07
	LSTM	99.92	1.28	99.99	0	99.82	0.05	0.13	6.94	595.81

further testing prior to deployment in real-world environments is needed.

In the future, it could be valuable to explore more DL models and their generalization capacity, as well as explainability methods, to further investigate the high and low FPR values and understand the reasons why models misclassify attacks and benign traffic. Additionally, testing more combinations of datasets could uncover a hidden pattern that could lead to improvement in model performance using other datasets. Moreover, adversarial training to help models more clearly distinguish between benign and malicious traffic could be pertinent. Since the class imbalance is so evident in GeNIS, aggregating it with other benign traffic datasets could provide models with a clearer distinction between benign and malicious traffic.

ACKNOWLEDGEMENTS

This research was supported by the SAFE project, which has received funding from the European Cybersecurity Competence Centre under grant agreement 101190370. Furthermore, the project UID/00760/2025 has also funded this work.

Daniela Pinto thanks Fundação para a Ciência e a Tecnologia (FCT) / Ministério da Educação, Ciência e a Inovação (MECI) for the financial support through a PhD Studentship (2024.18482.PRT).

REFERENCES

- Alshaibi, A., Al-Ani, M., Al-Azzawi, A., Konev, A., and Shelupanov, A. (2022). The comparison of cybersecurity datasets. *Data*, 7(2).
- Biswas, S. K., Dasari, A. K., Hossain, M. S., Rousanuzaman, Bhattacharya, A., Debasmita Saha, and Das, A. K. (2024). Performance of ensemble learning techniques for network-based intrusion detection system (nids): A comparative study. In Bansal, J. C., Borah, S., Hussain, S., and Salhi, S., editors, *Computing and Machine Learning*, pages 403–414, Singapore. Springer Nature Singapore.
- Boukraa, L., Essahraoui, S., Makkaoui, K. E., Ouahbi, I., Maleh, Y., and Esbaï, R. (2025). Enhancing ddos attack detection in software-defined networking: a comparative study of machine learning algorithms using benchmark datasets. *EDPACS*, 70(9):1–20.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1):5–32.
- Chen, T. and Guestrin, C. (2016). Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, KDD '16, page 785–794, New York, NY, USA. Association for Computing Machinery.
- Dixit, U. (2025). Enhancing intrusion detection performance with a hybrid module on kdd99 dataset. In *Computational Intelligence and its Applications*, pages 88–104. Bentham Science Publishers.
- Ferriyan, A., Thamrin, A. H., Takeda, K., and Murai, J. (2021). Generating network intrusion detection dataset based on real and encrypted synthetic attack traffic. *Applied Sciences*, 11(17).
- Fuhnwi, G. S., Revelle, M., and Izurieta, C. (2024). Improving network intrusion detection performance : An empirical evaluation using extreme gradient boosting (xgboost) with recursive feature elimination. In *2024 IEEE 3rd International Conference on AI in Cybersecurity (ICAIC)*, pages 1–8.
- Gao, J., Wang, Z., Jin, T., Cheng, J., Lei, Z., and Gao, S. (2024). Information gain ratio-based sub-feature grouping empowers particle swarm optimization for feature selection. *Knowledge-Based Systems*, 286:111380.
- Gong, Y., Liu, G., Xue, Y., Li, R., and Meng, L. (2023). A survey on dataset quality in machine learning. *Information and Software Technology*, 162:107268.
- Hnamte, V. and Hussain, J. (2021). An extensive survey on intrusion detection systems: Datasets and challenges for modern scenario. In *2021 3rd International Conference on Electrical, Control and Instrumentation Engineering (ICECIE)*, pages 1–10.
- Hochreiter, S. and Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8):1735–1780.
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., and Liu, T.-Y. (2017). Lightgbm: A highly efficient gradient boosting decision tree. In Guyon, I., Luxburg, U. V., Bengio, S., Wallach, H., Fergus, R., Vishwanathan, S., and Garnett, R., editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc.
- Kenyon, A., Deka, L., and Elizondo, D. (2020). Are public intrusion datasets fit for purpose characterising the state of the art in intrusion event datasets. *Computers & Security*, 99:102022.
- Khajgiwale, P. (2024). An essay on detailed performance assessment ensemble-based predictive modelling in network intrusion detection systems. In *2024 IEEE Students Conference on Engineering and Systems (SCES)*, pages 1–6.
- More, S., Idrissi, M., Mahmoud, H., and Asyhari, A. T. (2024). Enhanced intrusion detection systems performance with unsw-nb15 data analysis. *Algorithms*, 17(2).
- Moustafa, N. and Slay, J. (2015). Unsw-nb15: a comprehensive data set for network intrusion detection systems (unsw-nb15 network data set). In *2015 Military Communications and Information Systems Conference (MilCIS)*, pages 1–6.
- Pinto, D., Amorim, I., Maia, E., and Praça, I. (2024). A novel approach to network traffic analysis: the hera tool. In *2024 IEEE 23rd International Conference on*

- Trust, Security and Privacy in Computing and Communications (TrustCom)*, pages 1850–1856.
- Pinto, D., Vitorino, J., Maia, E., Amorim, I., and Praça, I. (2025). Flow exporter impact on intelligent intrusion detection systems. In *Proceedings of the 11th International Conference on Information Systems Security and Privacy - Volume 2: ICISPP*, pages 289–298. INSTICC, SciTePress.
- Rai, H. M., Pal, A., Ergash o'g'li, R. A., Kholmirezkhon Ugli, B. A., and Shokirovich, Y. S. (2025). Advanced ai-powered intrusion detection systems in cybersecurity protocols for network protection. *Procedia Computer Science*, 259:140–149. Sixth International Conference on Futuristic Trends in Networks and Computing Technologies (FTNCT06), held in Utarakhand, India.
- Reddy, L. S., Pasha, S. G., Bandela, H. B., Murthy, K. V. S. S., Naidu, U. G., and Shankar, R. S. (2024). Enhancing heart disease prediction with multiple imputation and feature selection in xgboost. In *2024 2nd International Conference on Advancement in Computation & Computer Technologies (InCACCT)*, pages 419–424.
- reddy, T. S. and Sathya, R. (2022). Ensemble machine learning techniques for attack prediction in NIDS environment. *ijcsm*, pages 78–82.
- Sarhan, M., Layeghy, S., and Portmann, M. (2022). Towards a standard feature set for network intrusion detection system datasets. *Mobile Networks and Applications*, 27(1):357–370.
- Setiawan, H., Faticah, C., and Saikhu, A. (2024). Feature selection for multilabel classification of student feedback using filter and metaheuristic methods. In *2024 International Conference on Intelligent Cybernetics Technology & Applications (ICICyTA)*, pages 1228–1233.
- Sharafaldin, I., Habibi Lashkari, A., and Ghorbani, A. A. (2018). Toward generating a new intrusion detection dataset and intrusion traffic characterization. In *Proceedings of the 4th International Conference on Information Systems Security and Privacy - ICISPP*, pages 108–116. INSTICC, SciTePress.
- Silva, M., Pinto, D., Vitorino, J., Gonçalves, J., Maia, E., and Praça, I. (2025a). Genis: A modular dataset for network intrusion detection and classification. *Data in Brief*, 60:111487.
- Silva, M., Vitorino, J., Maia, E., and Praça, I. (2025b). Efficient network traffic feature sets for iot intrusion detection. In Mehmood, R., Hernández, G., Praça, I., Wikarek, J., Loukanova, R., Monteiro dos Reis, A., Skarmeta, A., and Lombardi, E., editors, *Distributed Computing and Artificial Intelligence, Special Sessions I, 21st International Conference*, pages 3–13, Cham. Springer Nature Switzerland.
- Tariq, T. B., Khosa, S. N., Hadi, M. Z., Kiran, T., Mansab, M., Akram, U., and Mushtaq, M. F. (2025). Intelligent cyber security framework for threat detection using ensemble learning techniques. *Journal of Computing & Biomedical Informatics*, 8(02).
- Vibhute, A. D., Khan, M., Kanade, A., Patil, C. H., Gaikwad, S. V., Patel, K. K., and Saini, J. R. (2024). An lstm-based novel near-real-time multiclass network intrusion detection system for complex cloud environments. *Concurrency and Computation: Practice and Experience*, 36(11):e8024.
- Wang, Y., Liu, Y., Li, W., Deng, M., and Wang, K. (2025). Online estimation method for extreme learning machine with kernels based on the multi-innovation theory and intelligent optimization strategy. *ISA Transactions*, 156:142–153.
- Zha, C., Wang, Z., Fan, Y., Bai, B., Zhang, Y., Shi, S., and Zhang, R. (2025). A-nids: Adaptive network intrusion detection system based on clustering and stacked ctgan. *IEEE Transactions on Information Forensics and Security*, 20:3204–3219.